

Md. Estiak Rahman Ayon

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Portfolio | LinkedIn | GitHub

Professional Summary

Computer Science graduate with hands-on experience in Machine Learning, MLOps, Natural Language Processing, Computer Vision, and model deployment. Experienced in building end-to-end ML pipelines using DVC, MLflow, Docker, GitHub Actions, AWS, FastAPI, and Flask, covering experiment tracking, model versioning, CI/CD, and cloud deployment. Gained industry experience at Walton Hi-Tech Industries PLC, contributing to internal applications supporting manufacturing operations. Strong in Python, data processing, model evaluation, API development, and production-oriented ML workflows, with a focus on building reliable and scalable AI systems.

Education

Bachelor of Science in Computer Science and Engineering

American International University-Bangladesh (AIUB)

CGPA: 3.54 / 4.00

Sep 2022 – Aug 2026

Dhaka, Bangladesh

Higher Secondary Certificate (Science)

Shafipur Ideal Public School & College

GPA: 5.00 / 5.00

2021

Shafipur, Gazipur

Professional Experience

Software Engineer Intern

Walton Hi-Tech Industries PLC

Feb 2026 – Apr 2026

Gazipur, Bangladesh

- Built and maintained internal web applications supporting manufacturing operations using PHP, MySQL, JavaScript, and Bootstrap.
- Implemented role-based access control and approval workflows to manage department-specific records and operational processes.
- Developed responsive interfaces for applications used directly within manufacturing environments.
- Worked with production-line operators to collect requirements, understand workflow problems, and demonstrate newly released system features.
- Collaborated with operational teams during development, testing, and deployment of internal software improvements.

Projects

Bangladesh Electricity Demand Forecasting – End-to-End MLOps Project | GitHub | Live Demo

- Built a production-oriented electricity demand forecasting system using BPDB demand data and Open-Meteo weather data for national and regional predictions.
- Developed and evaluated time-series ML models with MLflow/DagsHub, achieving 4.89% mean CV MAPE and 5.68% holdout MAPE for national forecasting.
- Implemented automated data pipelines and versioning using DVC, GitHub Actions, and scheduled refresh workflows, including historical and multi-day forecast updates.
- Deployed the system with FastAPI, Streamlit, Docker, and Render, providing a public dashboard, REST API, historical evaluation, and extended demand forecasts.

Sentiment Analysis System with Automated MLOps Workflow | GitHub

- Built a reproducible end-to-end ML pipeline using DVC for data ingestion, preprocessing, feature engineering, model training, and evaluation.
- Integrated MLflow with DagsHub for experiment tracking and model registration, with dataset and artifact versioning using AWS S3.
- Containerized the Flask inference application using Docker and implemented automated build, testing, and AWS ECR image-push workflows using GitHub Actions.

Vehicle Insurance Prediction – End-to-End MLOps Pipeline | GitHub

- Developed a modular machine learning pipeline covering MongoDB data ingestion, validation, transformation, model training, and evaluation.
- Built a FastAPI-based model inference service and containerized the application using Docker.
- Integrated AWS S3 and ECR for model and container artifact management.
- Automated application build and deployment to AWS EC2 using GitHub Actions and a self-hosted runner.

Inventory Management System – Barcode-Based Product Tracking | GitHub

- Built a full-stack inventory system in PHP, MySQL and Bootstrap tracking products through their full lifecycle via unique barcodes.
- Implemented a five-role access system with stage-specific permissions and automatic stock in/out updates on each scan.
- Designed a normalised nine-table schema with activity logging, supporting stock, summary and barcode-wise history reports.

Restaurant Management System – Multi-Role Ordering Platform | GitHub

- Built a full-stack ordering platform in PHP and MySQL with separate admin, customer and rider interfaces.
- Implemented authentication, menu management, cart and order placement with AJAX-driven updates.
- Added delivery tracking allowing riders to update order status and customers to follow progress in real time.

Research Experience

Undergraduate Thesis – BanglaCXR: Vision-Language Pipeline for Bangla Chest X-Ray Report Generation

- Developed a two-stage vision-language pipeline for Bangla chest X-ray report generation using RAD-DINO, Phi-3.5 Mini, and 4-bit QLoRA on a single 8 GB GPU.
- Achieved 0.8882 F1-score and 0.9795 recall for abnormality detection on the IU X-Ray dataset.
- Integrated NLLB-200 with a medical glossary for Bangla translation; manuscript in preparation for journal submission.

Technical Skills

Programming: Python, SQL, C++, C#, JavaScript, PHP, R
Machine Learning & Data Science: PyTorch/TensorFlow, scikit-learn, pandas, NumPy, Matplotlib, Data pre-processing, Feature Engineering, Model Building & Evaluation
MLOps & Deployment: MLflow, DVC, DagsHub, Docker, Kubernetes, CI/CD, AWS EC2, AWS S3, AWS ECR, Model Versioning, Experiment Tracking
Backend Development: Laravel, Node.js, FastAPI, Flask, REST APIs,
Front-end Development: HTML, CSS, Bootstrap, Tailwind CSS, React.js, AJAX
Database: MySQL, MongoDB, Database Design
Tools: Git, GitHub, GitLab, Linux

Certifications

- **Data Science Job Simulation** – British Airways via Forage | [View](#)
- **Software Engineering Job Simulation** – Electronic Arts via Forage | [View](#)
- **Power BI for Beginners** – Microsoft via Simplilearn | [View](#)

Volunteering

BD Clean | [View](#) Nov 2023

- Participated in grassroots public cleanliness and environmental drives across Gazipur to encourage responsible civic waste management
- Collaborated with regional volunteer teams to coordinate community clean-up initiatives and promote sustainable hygiene practices

Recommendations

Aminun Nahar
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